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CHE 313 HW 1

① 1.01

②

$z_1 = 0.3 \text{ mol\% ethanol}$

③

$$F = 1200 \frac{\text{kmol}}{\text{h}}$$

$z_2 = 0.7 \text{ mol\% water}$

$$\text{Ethanol} = 1200 \frac{\text{kmol}}{\text{h}} \times 0.3 = 360 \frac{\text{kmol}}{\text{h}}$$

$$\frac{360 \text{ kmol}}{1 \text{ h}} \times \frac{46 \text{ kg}}{1 \text{ kmol}} = 16560 \frac{\text{kg}}{\text{h}}$$

$$\text{Water} = 1200 \frac{\text{kmol}}{\text{h}} - 360 \frac{\text{kmol}}{\text{h}} = 840 \frac{\text{kmol}}{\text{h}}$$

$$\frac{840 \text{ kmol}}{1 \text{ h}} \times \frac{18.016 \text{ kg}}{1 \text{ kmol}} = 15133 \frac{\text{kg}}{\text{h}}$$

$$16560 + 15133 = 31693 \frac{\text{kg}}{\text{h}}$$

② 1.D3

0.3 wt% ethanol
0.7 wt% water

$$F = 1200 \frac{\text{kmol}}{\text{h}}$$

$$Z_1(\text{ethanol mol frac}) = \frac{m \times 0.5}{46.1 \frac{\text{kg}}{\text{kmol}}} + \frac{m \times 0.7}{18.01 \frac{\text{kg}}{\text{kmol}}} = 1200 \frac{\text{kmol}}{\text{h}}$$

$$m(0.00651 + 0.0388) = 1200$$

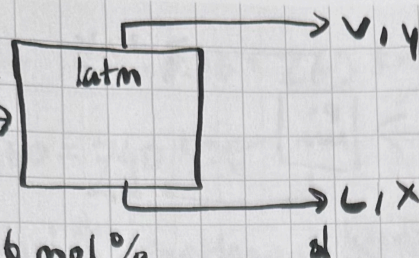
$$\hookrightarrow \text{Total Flow rate} = 26446 \frac{\text{kg}}{\text{h}}$$

$$n_E = 0.3(26446) / 46.07 = 172 \frac{\text{kmol}}{\text{h}}$$

$$\begin{matrix} \text{mol} \\ \text{frac} \end{matrix} \quad x_E = \frac{172}{1200} = 0.143$$

3

methanol: z_1
water: z_2



Equilibrium
Data: Fig 2.8

a)

$$z_1 = 0.6 \text{ mol}\%$$

$$V = 0.4 (F)$$

$$F = 100 \text{ kmol/h}$$

$$V = 0.4 (100) = 40 \text{ kmol/hr}$$

$$L = F - V = 60 \text{ kmol/hr}$$

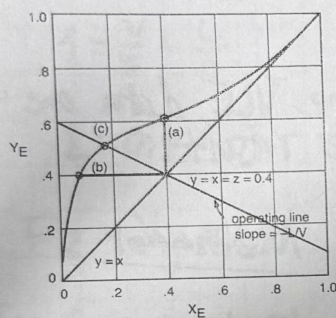


FIGURE 2-8. McCabe-Thiele diagram for binary flash distillation of ethanol-water; illustrated for Example 2-1

$$Fz = Vy + Lx$$

$$\rightarrow y = \frac{F}{V} z - \frac{L}{V} x \quad \text{Operating eqn}$$

$$y = mx + b$$

$$y = \frac{100}{40} (0.6) - \frac{60}{40} x$$

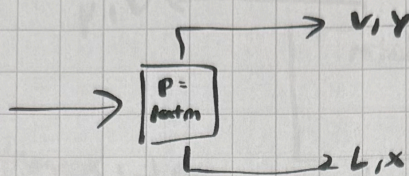
$$y = 1.5 - 1.5x$$

graphing this with the VLE data gives us

$$x = 0.49 \quad + \quad y = 0.77 \quad \text{for Methanol}$$

$$x = 0.51 \quad + \quad y = 0.23 \quad \text{for Water}$$

b) $F = 1500 \text{ kmol/h}$
 $\rightarrow z_1 = 0.6$
 $z_2 = 0.4$



$$V = 0.4(1500) = 600 \text{ kmol/h}$$

$$L = F - V = 900 \text{ kmol/h}$$

$$Fz = Vy + Lx \rightarrow y = \frac{F}{V}z - \frac{L}{V}x$$

$$y = \frac{1500}{600}(0.6) - \frac{900}{600}x$$

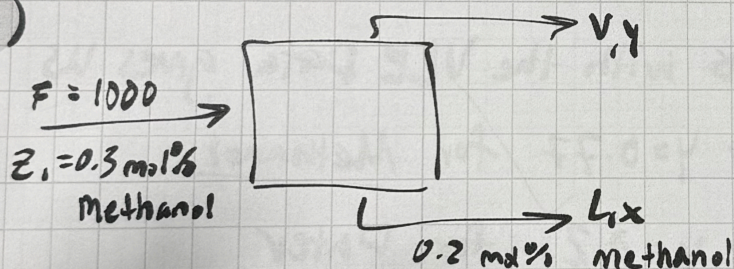
$$y = 1.5 - 1.5x$$

Since the operating function & the VLE data are the same as part a, we get the same results:

$$x = 0.49 \quad \& \quad y = 0.77 \quad \text{for Methanol}$$

$$x = 0.51 \quad \& \quad y = 0.23 \quad \text{for Water}$$

c)



$$\frac{V}{F} = ?$$

$$y = \frac{F}{V}z + \frac{L}{V}x \rightarrow y = \frac{F}{V}z + \frac{(F-V)}{V}x$$

$$\text{if } \frac{V}{F} = f \Rightarrow y = \frac{1}{f}z + \frac{1-f}{f}x \Rightarrow \text{known}$$

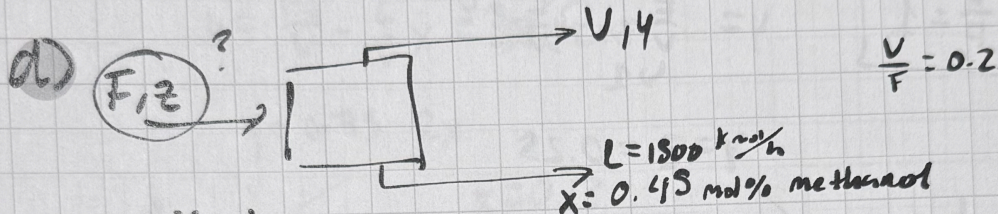
$$\left. \begin{array}{l} y = 0.579 \\ x = 0.2 \\ z = 0.3 \end{array} \right\} \text{VLE DATA}$$

$$0.579 = \frac{1}{f}(0.3) + \frac{1-f}{f}(0.2) \Rightarrow f = 0.26$$

$$V = F(0.26) = 1000 \times 0.26 = 260 \text{ kmol/h}$$

$$L = 1000 - 260 = 740 \text{ kmol/h}$$

$$\begin{cases} y = 0.26 \text{ mol\% methanol} \\ x = 0.2 \text{ mol\% methanol} \end{cases}$$



$$1 = \frac{V}{F} + \frac{L}{F}$$

$$1 = 0.2 + \frac{L}{F} \Rightarrow \frac{L}{F} = 0.8 = \frac{1500}{F} \quad F = 1875 \frac{\text{kmol}}{\text{h}}$$

$$Fz = Vy + Lx \quad V = F - L = 375 \frac{\text{kmol}}{\text{h}}$$

to find y we use VLE DATA + interpolate

$$\begin{array}{l} 0.4, 0.729 \\ 0.5, 0.779 \end{array}$$

$$y = y_1 + (x - x_1) \frac{(y_2 - y_1)}{(x_2 - x_1)} = 0.754$$

$$\text{SD, } F = 1875$$

$$V = 375$$

$$L = 1500$$

$$z = ?$$

$$y = 0.754$$

$$x = 0.45$$

$$Fz = Vy + Lx = (1875)(z) = (375)(0.754) + (1500)(0.45)$$

$$z = 0.51$$

④

25 mol%
methanol

1 atm

V, y

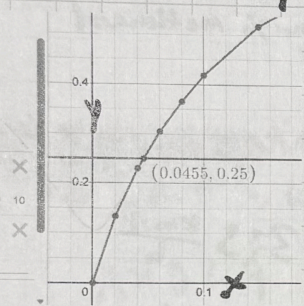
L, x

a) all the feed is vaporized

$$\frac{V}{F} = 1$$

$$y = \frac{F}{V} z - \frac{L}{V} x^0$$

$$y = z = 0.25$$



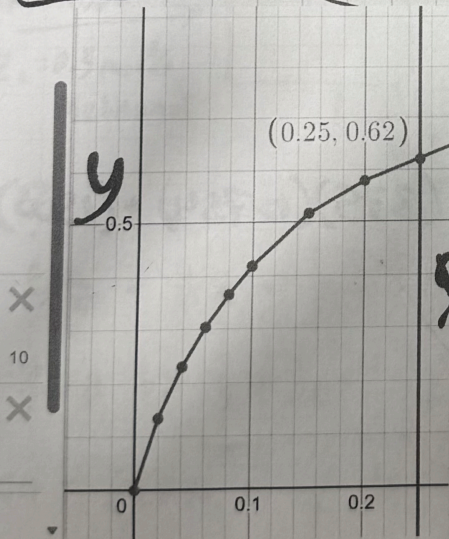
$$y = 0.25$$

$$x = 0.045$$

b) None of feed is vaporized

$$V = 0$$

$$Fz = V y^0 + L x \Rightarrow z = x = 0.25$$



$$x = 0.25$$

$$y = 0.62$$

c) $\frac{1}{3}$ of feed is vaporized

$$\frac{V}{F} = \frac{1}{3}$$

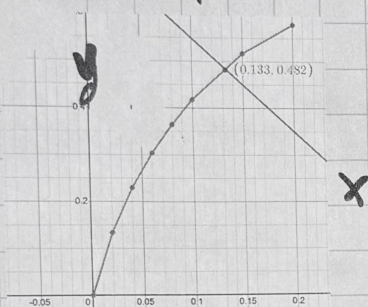
$$\frac{L}{F} = \frac{2}{3}$$

$$V = \frac{1}{3} \times F$$

$$L = \frac{2}{3} \times F$$

$$y = \frac{Fz}{V} - \frac{L}{V}x = 3z + 2x$$

$$y = 0.75 - 2x$$



$$x = 0.13$$

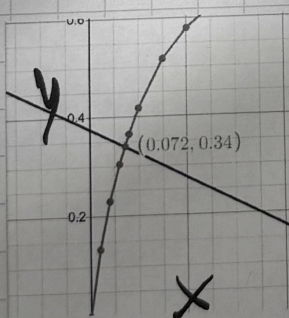
$$y = 0.48$$

d) $\frac{2}{3}$ of feed is vap

$$\frac{V}{F} = \frac{2}{3}, \quad \frac{L}{V} = \frac{1}{3}$$

$$y = \frac{Fz}{V} - \frac{L}{V}x$$

$$y = 0.375 - \frac{1}{2}x$$



$$x = 0.07$$

$$y = 0.34$$